MIL-H-29115A(SA)
3 January 1979
SUPERSEDING
MIL-H-29115(SA)
26 July 1972

MILITARY SPECIFICATION

HOOD, CHEMICAL, PROTECTIVE

This specification is approved for use by the Navy Supply Systems Command Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a protective hood made of an all cotton sateen material. The hood shall be of one type and size.

2. APPLICABLE DOCUMENTS

2.1 Issue of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-276	- Thread, Cotton
DDD-L-20	- Label; For Clothing, Equipage and Tentage (General Use)

MILITARY

MIL-C-10296	- Cloth, Sateen, Cotton
MIL-B-17757	- Boxes, Shipping, Fiberboard, (Modular Sizes)
MIL-T-43566	- Tape, Textile, Cotton, General Purpose, Natural or in Colors

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-751 - Stitches, Seams and Stitchings

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
*MIL-STD-1614 - Provisions for Evaluating Quality of Cloth Type Hoods

(Copies of specifications, standards, drawings and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* LAWS AND REGULATIONS

U. S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, D. C. 20036).

UNIFORM CLASSIFICATION COMMITTEE, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza., Chicago, Illinois 60606).

3. REQUIREMENTS

3.1 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample, in which case this specification shall govern.

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* 3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3).

3.3 Materials.

* 3.3.1 Basic material. The basic material shall be cotton, sateen cloth, shade Olive Green 107 conforming to class 1 of MIL-C-10296, with additional requirement that it shall show a minimum dynamic absorption of 60 percent when tested as specified in 4.4.1.

3.3.2 Thread, cotton. The cotton thread shall conform to type IA3, ticket numbers 30 and 50, 3 ply and ticket number 70, 2 ply of V-T-276. The thread shall be dyed Olive Drab S-1, C.A. 66022.

3.3.3 Tape. The tape for the tie-tapes shall be cotton, 1/2 inch wide, conforming to type I, class 3 of MIL-T-43566. The color shall be Olive Green 107.

* 3.3.4 Identification and Instruction. Each hood shall have a combination identification and instruction label conforming to classes 1 and 3 of DDD-L-20. The printing shall show fastness to laundering. The label shall bear the following inscription:

HOOD, CHEMICAL, PROTECTIVE
FIBER CONTENT: 100% Cotton
CONTRACT NO: DLA-100-00-0-0000 (Example)
STOCK NO: 8415-00-000-0000 (Example)
NAME OF SUPPLIER:

LAUNDER IN ACCORDANCE WITH PRESCRIBED REQUIREMENTS

1. Wear as a chemical agent protective head covering when a chemical attack is imminent.
2. Put on NAVY MK-V gas mask and adjust head straps for proper fit.
3. Draw the hood up over the head and insert the gas mask canisters through the face opening.
4. Tighten the back head adjustment tapes on hood, making sure a snug tight seal is maintained between the face opening and the mask to prevent gas penetration.
5. Secure bottom of hood to body by tieing shoulder underarms tie tapes.
6. Use Buddy system, if necessary, when donning hood.

DO NOT REMOVE THIS LABEL

3.4 Design. The hood shall consist of a four section head piece with an incorported flair skirt to drape over the shoulders providing protection below neck area. The hood shall have six tie tapes; four shoulder underarm tie tapes attached to bottom portion of skirt and two back headpiece adjustment tie tapes caught in side seaming of hood.

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3.4.1 Figure. Figure 1 is furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figure, the written specification shall govern.

3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government (see 6.3). The patterns allow 1/2 inch seam allowance for all double lapped and double stitched seams, except as otherwise indicated in Table I. The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way.

3.5.1 Pattern parts. The component parts shall be cut from the material as specified and in accordance with the following list of pattern parts:

Materials	Pattern Parts	Cut parts
Cloth, cotton sateen	Back	2
	Front	2
	Lining Front	2

3.6 Construction.

3.6.1 Stitches, seams and stitchings. Stitch, seam and stitching types as specified in Table I shall conform to the requirements of FED-STD-751. Whenever two or more methods for seams or stitching are given for the same operation, any one of them may be used. Where stitch type 401 is used, the looper (underthread) shall be on inside of hood. Seam allowance shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats or open seams will result. Edgestitching and topstitching shall be 1/16 inch from folded edge except as otherwise specified. Ends of seams produced by stitch type 401, shall be caught in other seams or stitchings.

3.6.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table I.

3.6.3 Thread breaks and ends of seams. Ends of all seams and stitchings when not caught in other seams or stitchings shall be backtacked not less than 3/8 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks (all stitch types shall be secured by stitching back of break not less than 1/2 inch. Skipped stitches or thread breaks (stitch type 401) may be repaired by using 301 stitch type.

3.7 Manufacturing operations requirements. The hoods shall be manufactured in accordance with operation requirements as specified in Table I. The contractor is not required to follow the exact sequence of operations provided the finished hoods are identical to that produced by following the sequence listed in Table I.

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Table I - Construction of hood

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
1.	Cutting a. Cut the hood in strict accordance with patterns, which shall show directional lines for cutting and instructions and notches for the proper assembly of all component parts.					
*	b. All component pattern parts shall be cut from one piece of material, except front lining which can be cut from ends and when cut from ends shall match the main assembles.					
*2.	Replacement of defective components. a. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in MIL-STD-1614 shall be removed from production and replaced with non-defective and properly matched components.					
3.	Join linings Join left front and right front linings with a double lapped and double stitched seam.	301 or 401	LSc-2	10-14	30 30	30 50
4.	Join fronts Join left front and right front pieces with a double lapped and double stitched seam.	301 or 401	LSc-2	10-14	30 30	30 50
5.	Join backs Join left back and right back pieces with a double lapped and double stitched seam.	301 or 401	LSc-2	10-14	30 30	30 50

NCTR Form 1, 15 Sep 1967

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Table I - Construction of hood (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
6.	Join lining to face opening					
	a. Stitch, face to face, front lining around face opening to front, 1/8 inch from raw edge.	301 or 401	SSa-1	10-14	30 30	30 50
	b. Turn lining and front to finished position. Fold under the bottom front raw edge of lining 1/4 (± 1/8) inch and stitch 1/16 inch from raw edge to front.	301 or 401	LSd-1	10-14	30 30	30 50
	c. Stitch, back to back, lining to front 3/16 inch from raw edge, along top and side seams.	301 or 401	SSa-1	10-14	30 30	30 50
7.	Make tie tapes					
	a. There shall be six tie tapes; the two back headpiece adjustment tapes shall measure 13 1/2 (+ 1/4) inches cut length and shall finish 12 1/2 ± 1/4 inches long and the four shoulder underarm tapes shall measure 17 ± 1/4 inches cut length and shall finish 16 ± 1/4 inches long.					
	b. Turn raw edge, of one end of tapes, 1/4 inch Fold the raw edge back over the body of the tape. Stitch or tack the turned and folded portion to body of tape.	301	EFb-1	10-14	30	30
8.	Join front and back					
	a. Position the two back headpiece adjustment tie tapes as indicated by marks on pattern in side seam. Tapes may be stay stitched.					

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Table I - Construction of hood (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
8.	Join front and back (cont'd)					
	b. Join the back, front and lining with a double lapped and double stitched seam with the backs over the fronts.	301 or 401	LSc-2	10-14	30 30	30 50
	c. Fold the two headpiece tie tapes over folded edges toward back and bartack or tack to side seams, full width of tape.	301 or bartack	LSa-1	10-14 28 per bartack	30 50	30 50
9.	Make hem and attach label					
	a. Fold hem under 1/8 inch and topstitch 1/16 inch from folded edge, simultaneously catching bottom of label in hem stitching.	301	EFb-1	10-14	30	30
	b. Stitch the remaining three sides of label to hood 1/16 inch from edge.	301	LSbj-1	10-14	30	30
	c. Hem shall finish 1/4 ± 1/8 inch wide and the label shall be centered over the center back seam.					
10.	Attach shoulder underarm tie tapes					
	a. Abut remaining raw edge to folded edge of hem as indicated by marks on pattern. Stitch tape to hood 1/4 inch from raw edge.	301	SSa-1	10-14	30	30
	b. Turn tape over line of stitching and box stitch to hood.	301	Similar LSd-1	10-14	30	30
11.	Clean hoods					
	a. Trim all ends and remove all loose threads.					
	b. Remove all spots and stains.					

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3.8 Measurements. The finished hood shall conform to the measurements shown in Table II (see figure 1). The measurements are expressed in inches.

Table II - Measurements of hood

	Measurement	Tolerance
A. Width of skirt	26	± 1/2
B. Length of face opening	7	± 1/4

A. Width of skirt. With the hood laid flat and smooth, measure from the intersection of side seam and hem to the same point on opposite side.

B. Length of face opening. With side seams superimposed lay the hood flat and smooth without stretch and measure from the top of the face opening to a level in line with the bottom of the face opening.

* 3.9 Workmanship. The finished hood shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable point value or defect limit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspections. The inspection requirements specified are classified as follows:

1. First article inspection (see 4.3)
2. Quality conformance inspection (see 4.4)

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* 4.3 First article inspection. When required (see 6.2) the first article submitted in accordance with 3.2 shall be visually inspected as specified in 4.4.2 for compliance with design, construction, workmanship, and dimensional requirements.

* 4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-1614.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended modified or qualified in this specification or applicable purchase documents. In addition to test methods specified in MIL-C-10296, the cloth shall be tested for dynamic absorption as specified in test method 5500 of FED-STD-191.

4.4.2 End item examination. Sampling and examination of the end item shall be performed in accordance with the provisions of MIL-STD-1614.

4.4.3 Packaging inspection. Packaging inspection shall be in accordance with MIL-STD-1614.

5. PACKING

5.1 Preservation-Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each hood shall be laid out flat, face up. The right and left skirt portions shall be folded inward to meet at the center seam of the hood. The hood shall then be folded in thirds from top to bottom to yield a measurement approximating 13 1/2 X 7 1/4 X 1/2 inches. Ten hoods shall be neatly stacked to form an even bundle and the resultant bundle shall be securely tied at each end with cotton tape or twine.

5.1.2 Level C, (commercial). Hoods shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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* 5.2.1 Level A. Ninety hoods, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed, and reinforced conforming to class weather-resistant, grade V2s, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class weather-resistant, variety DW, grade V15c or MIL-B-17757. Level A packages shall be packed flat, three in length, one in width, and three in depth within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade draft paper.

* 5.2.2 Level B. Ninety hoods, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to class domestic, grade 275, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic, variety DW, grade 200 of MIL-B-17757. Level A packages shall be packed flat, three in length, one in width, and three in depth within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with MIL-B-17757 and closed in accordance with the appendix of the box specification.

* 5.2.3 Level C (Commercial). Hoods, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that used by the contractor for retail distribution. Containers shall be in accordance with U. S. Postal Service Manual, Uniform Freight Classification Rules, or National Freight Classification Rules, as applicable.

5.4 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The hoods are intended to be impregnated with a special compound and worn in conjunction with Navy MARK V gas mask and other impregnated clothing items as part of a chemical protective ensemble.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When first article sample is required (see 3.2).
c. Selections of applicable levels of packaging and packing (see 5.1 and 5.2).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

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6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

* 6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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User activity:
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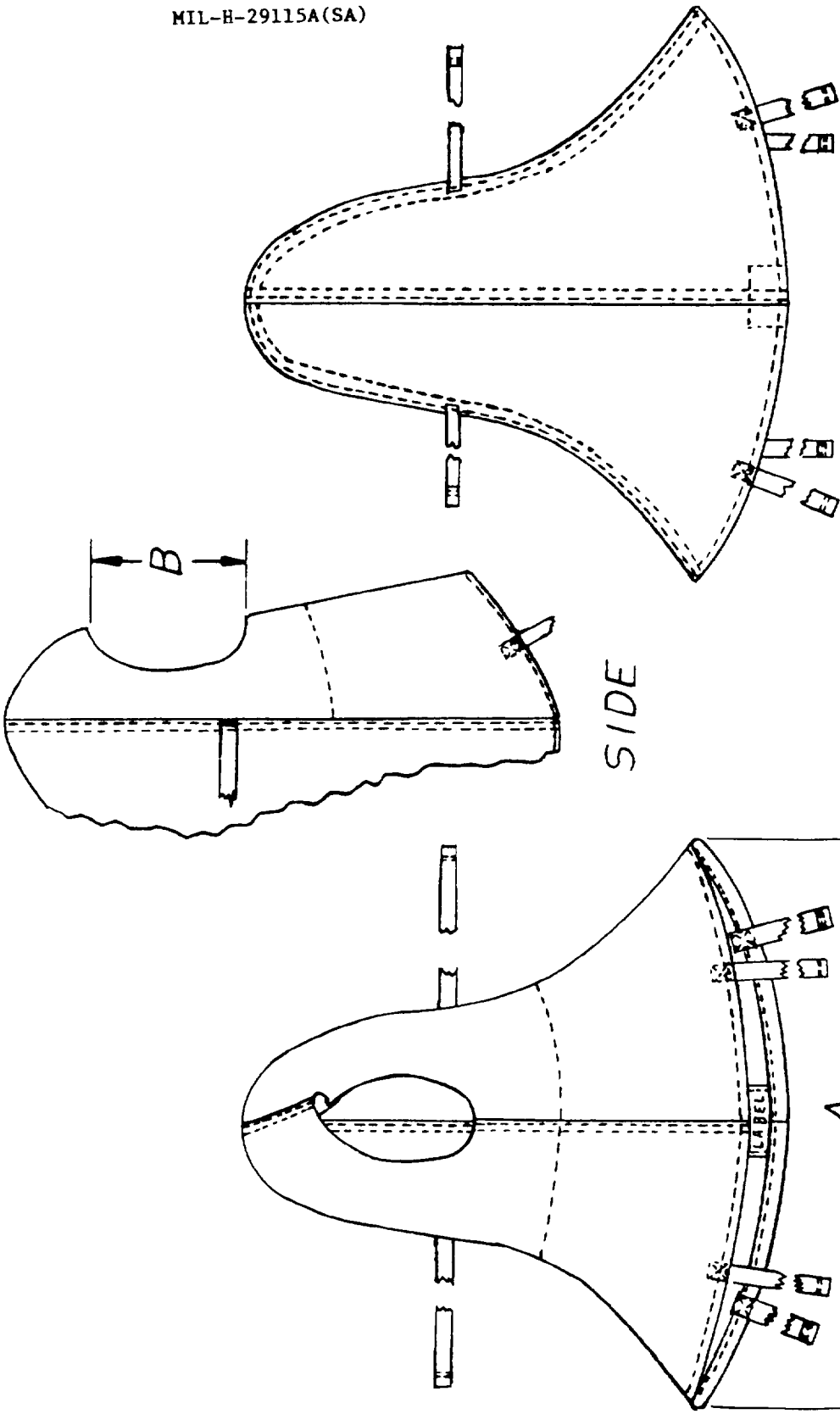


FIGURE 1 HOOD, CHEMICAL PROTECTIVE

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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